Data and Programming II: Final Project

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Background and Research Motivation

As of 2023, many U.S. counties continue to face significant gaps in healthcare capacity, while chronic disease burdens remain persistently high. Programs like the National Health Service Corps (NHSC) try to address this by placing clinicians in underserved areas through scholarships and loan repayment incentives. But with limited placements available, these are exactly the kinds of questions that matter for programs like NHSC and motivate our analysis. Therefore, our research questions include: where does additional supply actually matter most, and if resources are directed toward certain counties, how large is the expected impact, and does that look different depending on a county's starting supply conditions?

Data and Methodology

We combined four county-level datasets to build as comprehensive a picture as possible of each county's health burden, clinical capacity, and demographic baseline. (1) **Health Burden Estimates (from CDC):** This dataset provides estimates of chronic disease prevalence, related health measures and adult populations at U.S. county level. (2) **Health Workforce and Facility Capacity (from HRSA):** This dataset includes physician counts by specialty and hospital bed capacity at the county level. (3) **Population and Demographic Baselines (from U.S. Census / ACS 5-year estimates):** This dataset provides median income, which we used as control to make our prevalence predictions more robust. (4) **Geographic Data (from U.S. Census TIGER):** This dataset provides county-level geographic boundary data.

Data Preprocessing

The raw data files were large and complex, so we did two main things to make them workable. First, we streamlined each file by removing irrelevant or redundant fields, and keep only the core county-level information we needed. For disease outcomes, we focused on three representative conditions: diabetes as a primary care condition, coronary heart disease (CHD) as a specialty-intensive condition, and depression as a mental health condition with distinct supply dynamics. These were chosen to be illustrative and clear. On the supply and demographic side, we retained adult population baselines, county identifiers, overall physician availability, the most relevant specialties, hospital bed capacity, and median income as basic socioeconomic control. Once the files were trimmed down, we merged all four datasets at county level using county identifiers to produce a single unified dataset for analysis. Second, we created a set of analysis-ready variables to support our visualization and regressions. We converted raw prevalence figures into population-weighted county estimates and standardized all supply measures into "per 10,000 adults (18+)" units for comparison. We also built supply tiers by quintiles and an additional "No Providers" group (No providers -> High) to clearly capture the pattern of different resource-level counties.

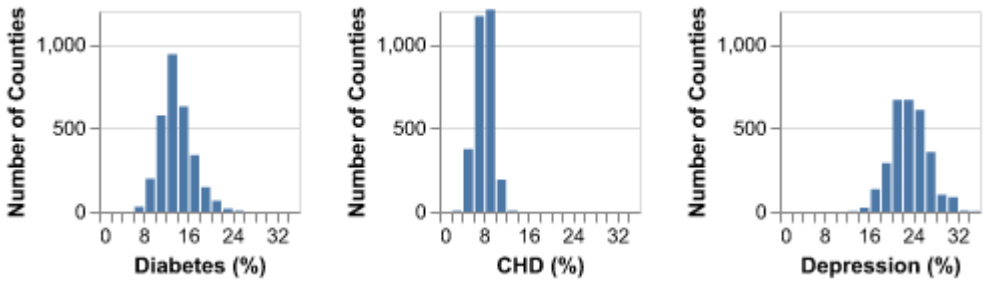
Approach

We start with static visualizations to show county-level patterns in disease burden and healthcare supply across the U.S., along with correlations between them. From there, we move into a dynamic & interactive Streamlit dashboard for users, mainly policy makers at programs like NHSC. Using selectbox and sliders, they can adjust different supply allocation scenarios for each disease case and see the estimated direction and magnitude of impact for set of counties.

Descriptive Patterns

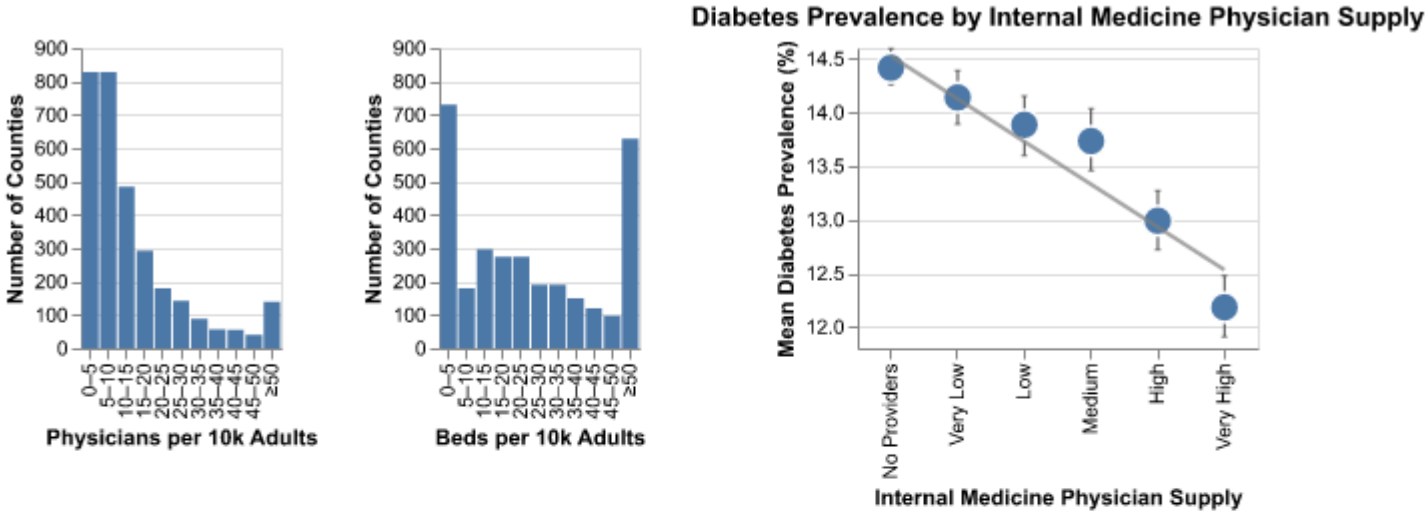
The diseases show different prevalence distributions across counties after excluding counties with zero prevalence (there's no NA in raw data).

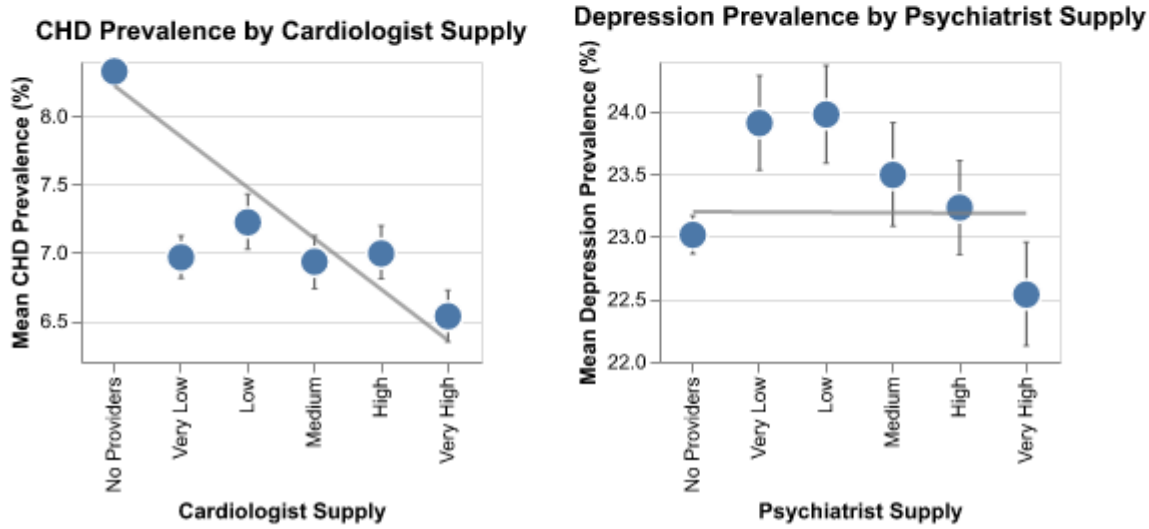
(1) Disease Burden: - CHD has the lowest overall prevalence but is the most concentrated, depression has the highest prevalence but is the most dispersed, and diabetes in between. These differences reflect both the underlying epidemiology of each condition and how each tends to be diagnosed. - CHD is a clearly defined, organically-based chronic condition that affects a smaller share of population. As a result, most counties cluster around similar rates with low prevalence and low spread. Depression works very differently. Whether someone gets diagnosed depends heavily on their willingness to seek care and access to mental health resources, none of which are evenly distributed across counties. The occurrence of mental health conditions is also higher than severe conditions. These provide a possible explanation for a much wider spread. Diabetes is meaningfully shaped by lifestyle and socioeconomic factors , but it also has fairly standardized screening and diagnostic criteria. That combination to some extent, is associated with moderate county-level variation — more dispersed than CHD, but more structured than depression.



(2) Supply: Counties with zero physician or bed supply are retained because zero supply itself reflects meaningful healthcare access constraints. Both physician and hospital bed supply are right-skewed: most counties cluster at lower supply levels, while a small number of counties have extremely high capacity. For visualization purposes, values above 50 per 10,000 adults are grouped into a single top category.

(3) Relationship Between Supply and Disease Prevalence: Then we look at how supply relates to prevalence. To make this easier to see, we grouped counties into six supply tiers — a "No Providers" group (zero supply) and five quintiles among non-zero counties, ranging from Very Low to Very High. The results vary across diseases. In the case of diabetes, the trend is clear. As the supply of internal medicine physicians increases, the prevalence of diabetes decreases. CHD also shows a general downward trend as cardiologist supply rises, but the middle supply tiers exhibit very similar prevalence levels, with only small differences across these groups and clearer separation mainly at the extremes. For depression, there is no clear trend, with considerable overlap among the supply tiers and only a weak overall trend.

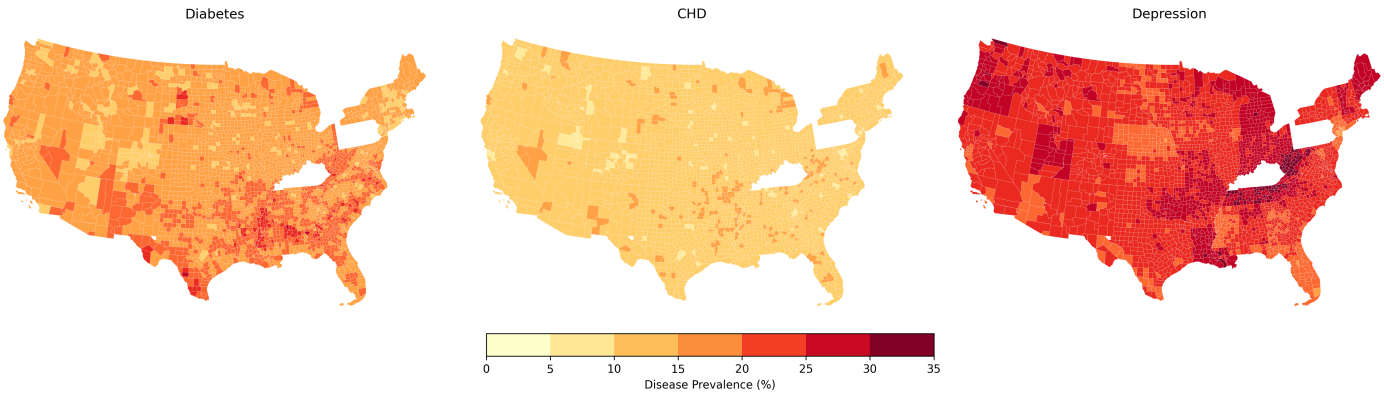


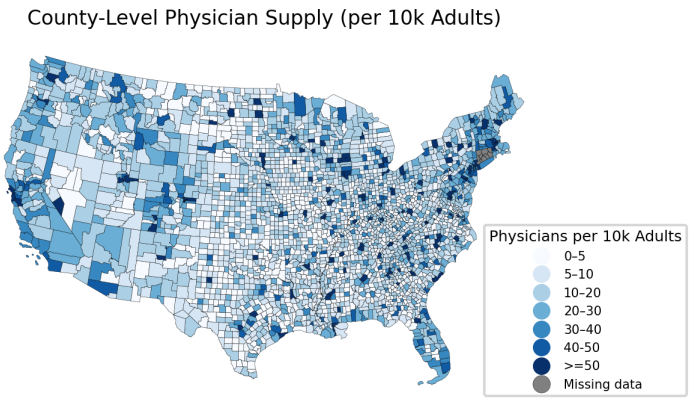


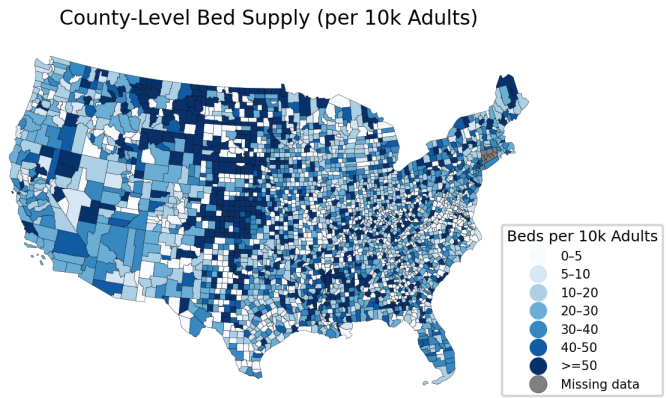
(4) Geographical Distribution

Clear regional patterns are visible across the US. Regarding diseases, counties with a prevalence of 0 are displayed as blank on the graph. Diabetes shows particularly high prevalence in parts of the South, while CHD also displays somewhat higher values in several southern counties. Depression, while more widespread geographically, is still regionally uneven rather than uniformly distributed. On the supply side, despite a few counties with missing data, the geographic disparities are striking. Physician

density and hospital bed capacity vary substantially across counties. Many large metropolitan areas tend to have a relatively high supply of physicians and hospital beds, while many smaller or less populated counties tend to have fewer supplies, though there are some that do not fit this pattern.







Streamlit Dashboard: https://final-project-shuyan-ziye-6pkom6vbzap9avqyw3m3ek.streamlit.app/

- Method: We use OLS regression to quantify the relationship between disease burden and healthcare supply, which then feeds into the scenario simulations. For each disease, we estimate a linear relationship between county-level prevalence y_i and the corresponding specialty physician supply S_i, controlling for observable county characteristics including adult population size (18+), hospital bed capacity per 10,000 adults, and median household income — to improve precision and reduce bias from observable differences across counties.
- How It Works: On the left panel, policy makers can select from three disease type (selectbox) and change supply level of the six supply tiers (slider). The dashboard then dynamically updates: the estimated change in health prevalence for counties in the selected supply tier, and the overall population-weighted prevalence across all selected counties. This allows users to directly compare the expected direction and magnitude of impact from reallocating resources to counties at different starting supply levels.

Key Findings

Disease burden and healthcare supply are shaped by many factors. Because of this complexity, our analysis does not attempt to establish causal relationships. Instead, the dashboard serves as an exploratory simulation tool that allows policy makers to examine how reallocating healthcare resources across counties with different starting conditions may influence predicted disease prevalence.

Across the three diseases, the simulation results show that the relationship between physician supply and predicted prevalence is not consistent across diseases or supply tiers. In the case of diabetes, the simulation results show that increasing the supply of internal medicine physicians is generally associated with decreasing predicted prevalence in most counties. The magnitude of the decrease is larger in the middle supply tiers. For CHD, the simulation results show that the relationship is more complex. Predicted prevalence increases in some of the middle supply tier counties but remains relatively stable or decreases slightly in the counties without cardiologists as well as those in the higher supply tiers. The simulation results for the case of depression show the most varied pattern. Predicted prevalence decreases in some counties but increases in others.

Several mechanisms could contribute to these patterns. In areas with very limited medical resources, increases in physician supply may improve access to care, earlier treatment, and continuity of disease management, which could contribute to lower observed prevalence. At the same time, increases in provider availability may also lead to more screening, diagnosis, and documentation, which can reveal cases that previously went undetected and therefore increase measured prevalence. The observed patterns likely reflect a combination of these mechanisms, and the underlying dynamics remain complex.